

Connect a Web App to Amazon Aurora

Melvin green

ID	NAME	ADDRESS
1	John Doe	123 Coding Ave, NJ
2	Testing It	29 July st, 2026

Introducing Today's Project!

What is Amazon Aurora?

Amazon Aurora is a relational database, and it is useful because it allows me to organize data into structured tables with defined relationships between them, making it easier to store, query, and manage large amounts of data efficiently. As a managed AWS service, Aurora also

handles tasks like backups, scaling, and maintenance automatically, which reduces the operational overhead compared to managing a database server manually.

How I used Amazon Aurora in this project

In today's project, I used Amazon Aurora to store data collected from the website and used the MySQL CLI to query and verify that data directly from the terminal. This project demonstrates my ability to connect a web application to a relational database, handle data storage and retrieval, and troubleshoot common issues involving permissions, SSH connections, and server configuration along the way.

One thing I didn't expect in this project was...

One thing I didn't expect from this project was learning how to connect my EC2 instance to my Linux terminal environment. This gave me valuable practice navigating the terminal using various commands, creating files, and writing scripts directly within those files. Overall, this project helped strengthen my comfort level working in a command line environment, a skill that extends well beyond this specific project.

This project took me...

This project took me about two and a half hours to complete. I took my time navigating through the CLI to make sure I understood each step along the way.

```
phantom@DESKTOP-M4G8VIB:~$ ssh -i ~/melsAuroraApp.pem ec2-user@18.227.83.252
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

      #_
     ~\  #####_          Amazon Linux 2023
    ~~ \  #####\
    ~~  \###|
    ~~   \#/  ____
    ~~    V~'  '->
        ~~~
         ..
        _/_/_/_/_
       _/m/'

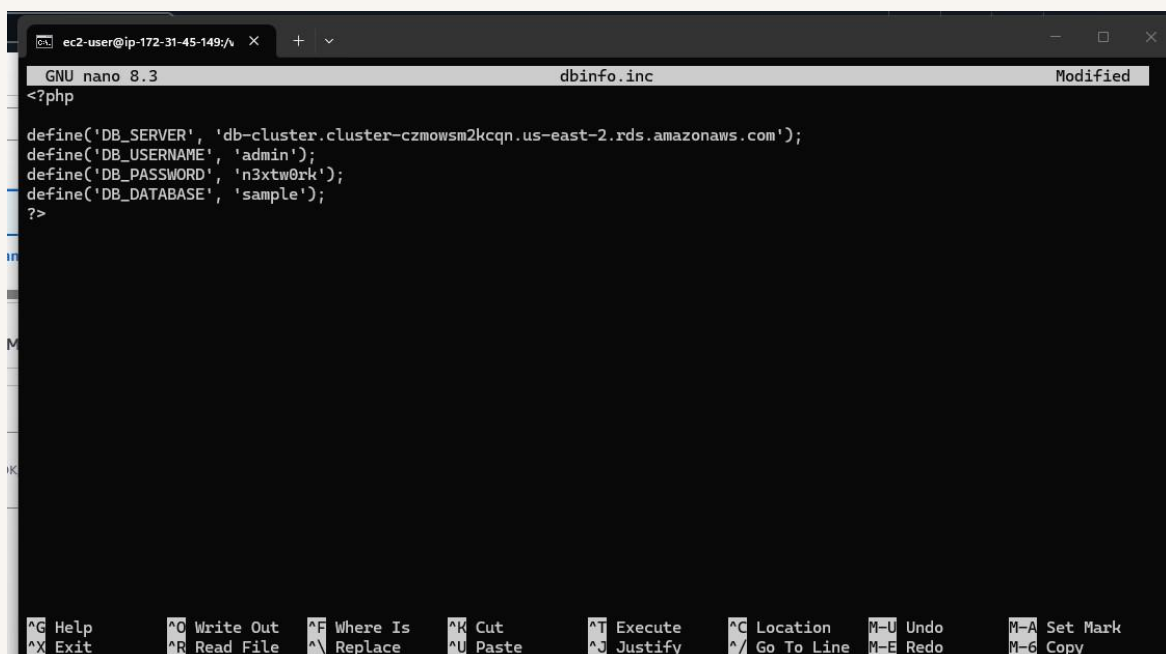
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-45-149 ~]$ |
```

To build my web app, I began by ensuring all software packages are up to date, running `sudo dnf update -y` to apply the latest security patches and updates. I then install the core components of a LAMP-style stack: the Apache web server to host the application, PHP to handle server side logic, `php-mysqli` to allow PHP to communicate with the database, and MariaDB as the relational database management system. This setup forms the foundation needed to run a dynamic, database driven web application on my EC2 instance.

Connecting my Web App to Aurora

I set up my EC2 instance's connection details to my database by first granting myself owner permissions. Since I am connected to my EC2 instance, which has admin access, I change the ownership by running `sudo chown ec2-user:ec2-user /var/www`, giving myself owner permissions over that directory. I then create a subfolder named `inc` and a new file within it named `dbinfo.inc`, which will store my database connection credentials. I edit that file using `nano`, a lightweight text editor designed to run entirely within the terminal. Within the file, I add the Aurora database endpoint details needed to establish the connection. Finally, I save the file and exit back to the previous directory.



```
ec2-user@ip-172-31-45-149:/var/www$ nano dbinfo.inc
GNU nano 8.3 dbinfo.inc Modified
<?php
define('DB_SERVER', 'db-cluster.cluster-czmowsm2kcqn.us-east-2.rds.amazonaws.com');
define('DB_USERNAME', 'admin');
define('DB_PASSWORD', 'n3xtw0rk');
define('DB_DATABASE', 'sample');
?>
```

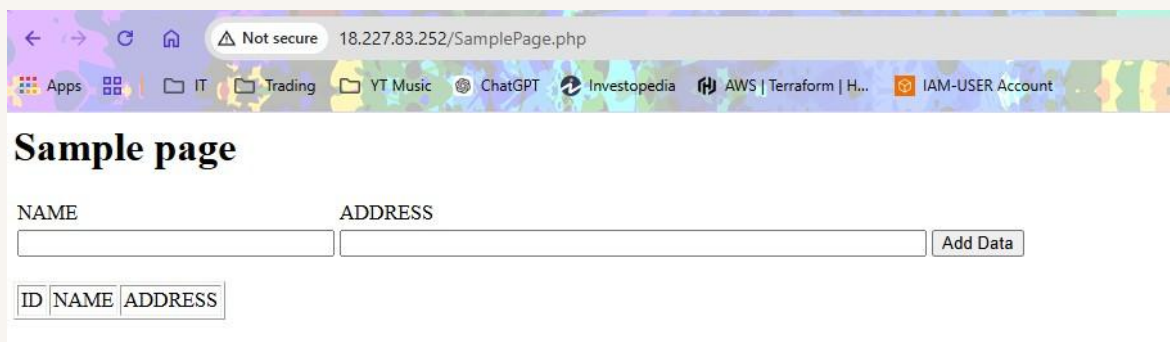
The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named `dbinfo.inc`. The code in the file is a PHP snippet that defines database connection details for an Aurora database. The details include the server name, username, password, and database name. The code is as follows:

```
<?php
define('DB_SERVER', 'db-cluster.cluster-czmowsm2kcqn.us-east-2.rds.amazonaws.com');
define('DB_USERNAME', 'admin');
define('DB_PASSWORD', 'n3xtw0rk');
define('DB_DATABASE', 'sample');
```

The terminal window also shows the nano editor's status bar at the bottom, which includes various keyboard shortcuts for editing and saving the file.

My Web App Upgrade

Next, I upgrade my web app by navigating to my html folder. From there, I create a new file named samplepage.php within the html directory and edit it using nano. I add a block of PHP and HTML code to improve the page's structure and styling, resulting in a more polished, user-friendly interface for the web application.



Testing my Web App

To make sure my web app was working correctly, I entered sample information into the website's form and submitted it. I then returned to the terminal, accessed the database using the MySQL CLI, and queried the relevant table to confirm that the data I submitted through the website had been successfully stored in the Aurora database.

```

Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 433
Server version: 8.0.42 6252a59a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sample |
| sys |
+-----+
5 rows in set (0.002 sec)

MySQL [(none)]> USE sample;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [sample]> SHOW TABLES;
+-----+
| Tables_in_sample |
+-----+
| EMPLOYEES |
+-----+
1 row in set (0.001 sec)

MySQL [sample]> DESCRIBE EMPLOYEES;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| ID    | int unsigned  | NO   | PRI | NULL    | auto_increment |
| NAME  | varchar(45)   | YES  |     | NULL    |                |
| ADDRESS | varchar(90)  | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

MySQL [sample]> SELECT * FROM EMPLOYEES;
+-----+-----+-----+
| ID | NAME      | ADDRESS          |
+-----+-----+-----+
| 1  | John Doe  | 123 Coding Ave, NJ |
| 2  | Testing It | 29 July st, 2026 |
+-----+-----+-----+

```